

bioassist.ai

Platform Specification — UX, Data, Recommendations, Integrations, and Business Model

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What this document contains

A build-ready blueprint for a clean, Material-inspired health platform that ingests documents + devices, normalizes data with provenance, and produces evidence-linked, measurable recommendations — with safety and privacy as first-class features.

- Wireframe-level page layouts and components
- Canonical data model (FHIR-aligned) + time-series schema + provenance
- Ingestion and normalization pipelines for PDFs/images/receipts/devices/EHR
- Recommendation engine design, guardrails, and a starter recommendation library
- Integration plan (HealthKit/Health Connect first, then wearables + labs + EHR)
- Monetization strategy, tiers, paywalls, and roadmap

Design goals: zero-friction start, deep capability on demand, user-owned data, and recommendations that are measurable and defensible.

Table of Contents

Table of Contents	2
1. Product Definition and Positioning	6
1.1 Target users and “jobs to be done”	6
1.2 Differentiators	6
2. UX and Information Architecture	7
2.1 Core UX principles	7
2.2 Navigation model (mobile + web)	7
2.3 The “empty state” strategy	7
3. Screen Specifications (Wireframe Level)	8
4. Data Ingestion, Normalization, and Provenance	12
4.1 Supported input sources (MVP → v1 → v2)	12
4.2 Document ingestion pipeline	12
4.3 Device ingestion pipeline	12
4.4 Normalization rules (non-negotiable)	13
5. Canonical Data Model	14
5.1 Core entities (minimum viable)	14
5.2 Provenance primitive (SourceSpan)	16
5.3 Storage choices (pragmatic defaults)	16
6. Architecture and API Surface	17
6.1 Services (suggested boundaries)	17
6.2 Internal contracts: events, not tight coupling	17

- 6.3 Public API (user-owned export) 17
- 6.4 Observability and quality metrics 18
- 7. Recommendation Engine and Safety Guardrails 19**
 - 7.1 Layered recommendation stack 19
 - 7.2 Recommendation contract (canonical JSON schema) 20
 - 7.3 Ranking and “priority stack” algorithm 20
 - 7.4 Safety guardrails (minimum set) 21
 - 7.5 Starter recommendation library (first release) 21
- 8. Evidence Map and Research System 22**
 - 8.1 Evidence map data model 22
 - 8.2 Research pipeline (human-reviewed) 22
 - 8.3 “No-citation, no-claim” rule for scientific statements 22
 - 8.4 Two explanation modes 22
- 9. Integrations Roadmap 23**
 - 9.1 Integration philosophy 23
 - 9.2 Priority matrix 23
 - 9.3 Device connection UX requirements 23
- 10. Privacy, Security, and Trust 24**
 - 10.1 Privacy UX primitives 24
 - 10.2 Security architecture (baseline) 24
 - 10.3 Compliance posture (practical) 24
- 11. Monetization and Business Model 25**
 - 11.1 Value-based packaging (tiers) 25
 - 11.2 Paywall philosophy 25

- 11.3 Revenue levers beyond subscriptions 25
- 11.4 Unit economics checklist 25
- 12. Growth, Retention, and Engagement Loops 26**
 - 12.1 Retention loops (recommended) 26
 - 12.2 Acquisition channels (practical) 26
 - 12.3 Avoid these engagement anti-patterns 26
- 13. Roadmap and Build Plan 27**
 - 13.1 Milestone roadmap 27
 - 13.2 MVP definition (must ship) 27
 - 13.3 Team and responsibilities (minimum viable team) 27
 - 13.4 Quality gates 27
- 14. Analytics and Product Instrumentation 28**
 - 14.1 Key product metrics 28
 - 14.2 Event taxonomy (starter set) 28
 - 14.3 Feedback loops 28
- 15. AI/LLM Implementation Notes 29**
 - 15.1 Where LLMs are appropriate 29
 - 15.2 Where LLMs are not appropriate (or must be heavily constrained) 29
 - 15.3 Prompting patterns (recommended) 29
 - 15.4 Prompt injection and document safety 29
- 16. Appendix: API Surface and Contracts (Starter) 30**
 - 16.1 Example REST endpoints 30
 - 16.2 UI query model (recommended pattern) 30

16.3 Caching strategy 30

17. Build Checklist 31

17.1 UX component checklist (Material-inspired) 31

17.2 Backend epics 31

17.3 Launch checklist 31

1. Product Definition and Positioning

What BioAssist is, who it is for, and why it wins.

BioAssist is a user-owned health operating system: it aggregates messy health data (documents, wearables, labs, EHR), turns it into a clean longitudinal record with provenance, and produces prioritized, evidence-linked recommendations that are measurable.

1.1 Target users and “jobs to be done”

Design for a spectrum. The platform must be useful with almost no data, but become more powerful as users connect sources.

- **Everyday optimizers:** want simple guidance and habit loops (sleep, activity, nutrition).
- **Condition managers:** want monitoring, trend detection, and preparation for clinician visits.
- **High performers:** want training load + recovery + biomarkers; experiment workflows.
- **Caregivers:** want a shared timeline and reminders; low cognitive load.
- **Clinicians (optional):** want patient packets (timeline + key metrics + questions).

1.2 Differentiators

- **Provenance-first data model:** every claim ties back to a source (device, PDF, EHR).
- **Recommendation contract:** every recommendation includes trigger, expected effect range, risks, cost/burden, and measurement plan.
- **Progressive disclosure UX:** minimal surface area until complexity is needed.
- **Closed-loop measurement:** recommendations are experiments with re-measurement.
- **Trust primitives:** audit logs, exportability, and transparent AI behavior.

Product “north star” metric

Increase the user’s **Actionable Signal** per week: the number of prioritized actions that are (1) evidence-backed, (2) feasible, and (3) measurable, that the user actually completes.

Secondary: retention (weekly active), successful data ingestions, and clinician packet exports.

2. UX and Information Architecture

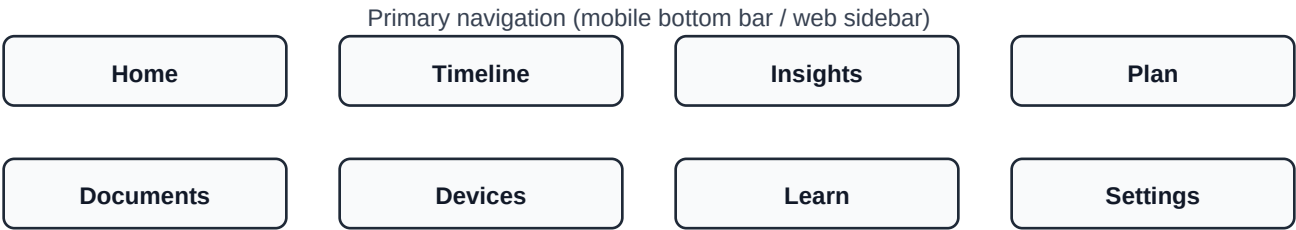
Material-inspired: clean surfaces, consistent components, minimal cognitive load.

2.1 Core UX principles

- **Start instantly:** user gets value in under 2 minutes (connect HealthKit/Health Connect or upload 1 PDF).
- **Progressive disclosure:** show 3–7 high-priority items; hide the rest behind “See details”.
- **One global search:** search across labs, documents, meds, conditions, and recommendations.
- **Explainability by default:** every chart and rec has “What is this?” + “Why do we think this?”.
- **Correctability:** user can fix extracted data; corrections become training data (with consent).
- **Non-alarming tone:** red flags only when clear; otherwise use neutral language.

2.2 Navigation model (mobile + web)

Keep the primary navigation stable. Everything else is drill-down inside these pages.



- **Mobile:** bottom navigation with 5 items (Home, Timeline, Plan, Documents, More). “More” opens Insights, Devices, Learn, Settings.
- **Web:** left sidebar with all primary pages visible.
- **Search:** persistent in app bar; typing shows chips for common filters.

2.3 The “empty state” strategy

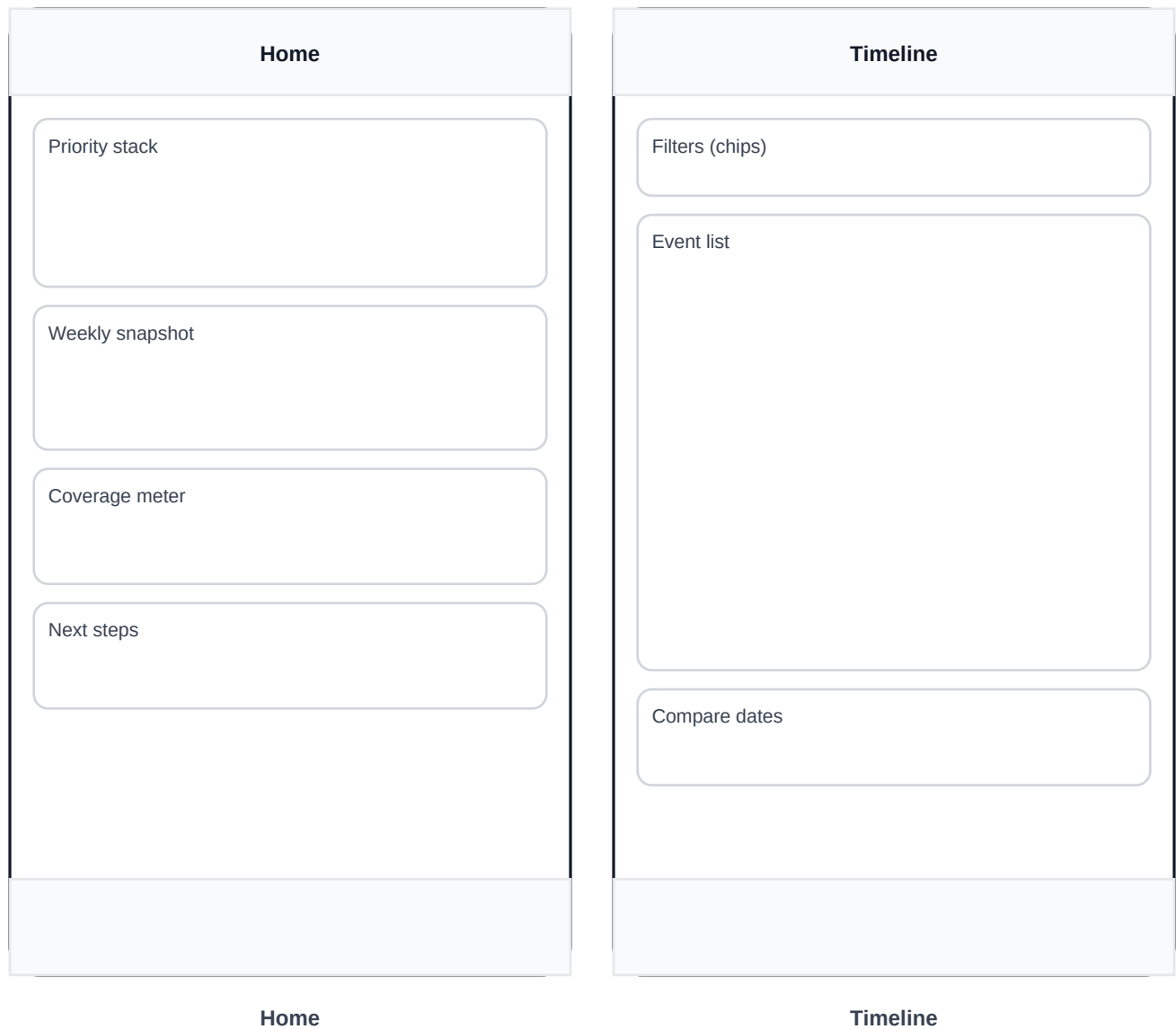
Empty states are the product. They should teach, not nag.

- Home empty: show 2 quick actions — **Connect device** and **Upload a document**.
- Timeline empty: show example timeline items and what will appear.
- Insights empty: show coverage checklist by domain (metabolic, sleep, cardio).
- Plan empty: show the first 3 recommendation templates (Sleep regularity, Daily walk, BP baseline).

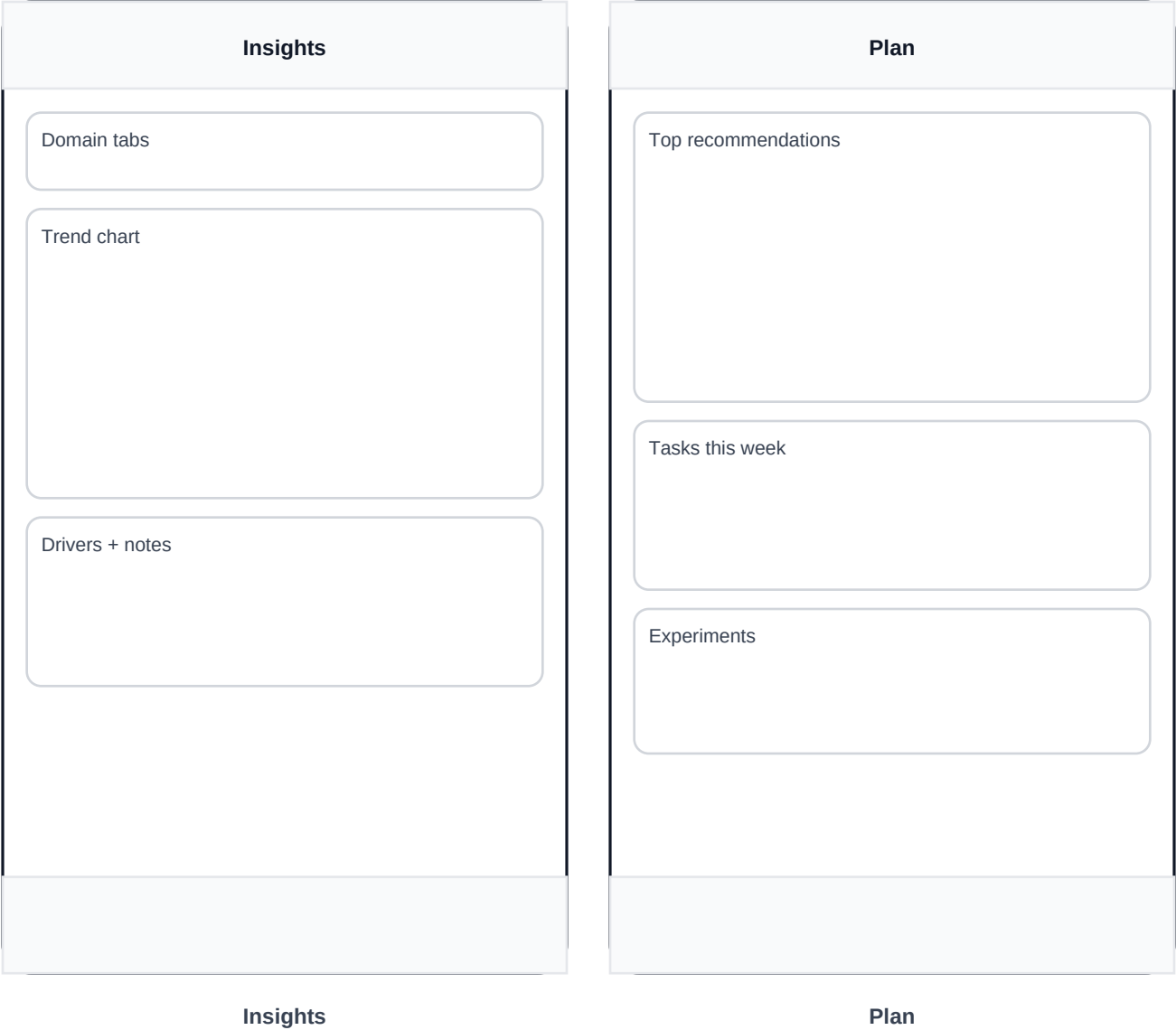
3. Screen Specifications (Wireframe Level)

What each page contains, how it behaves, and what the default view looks like.

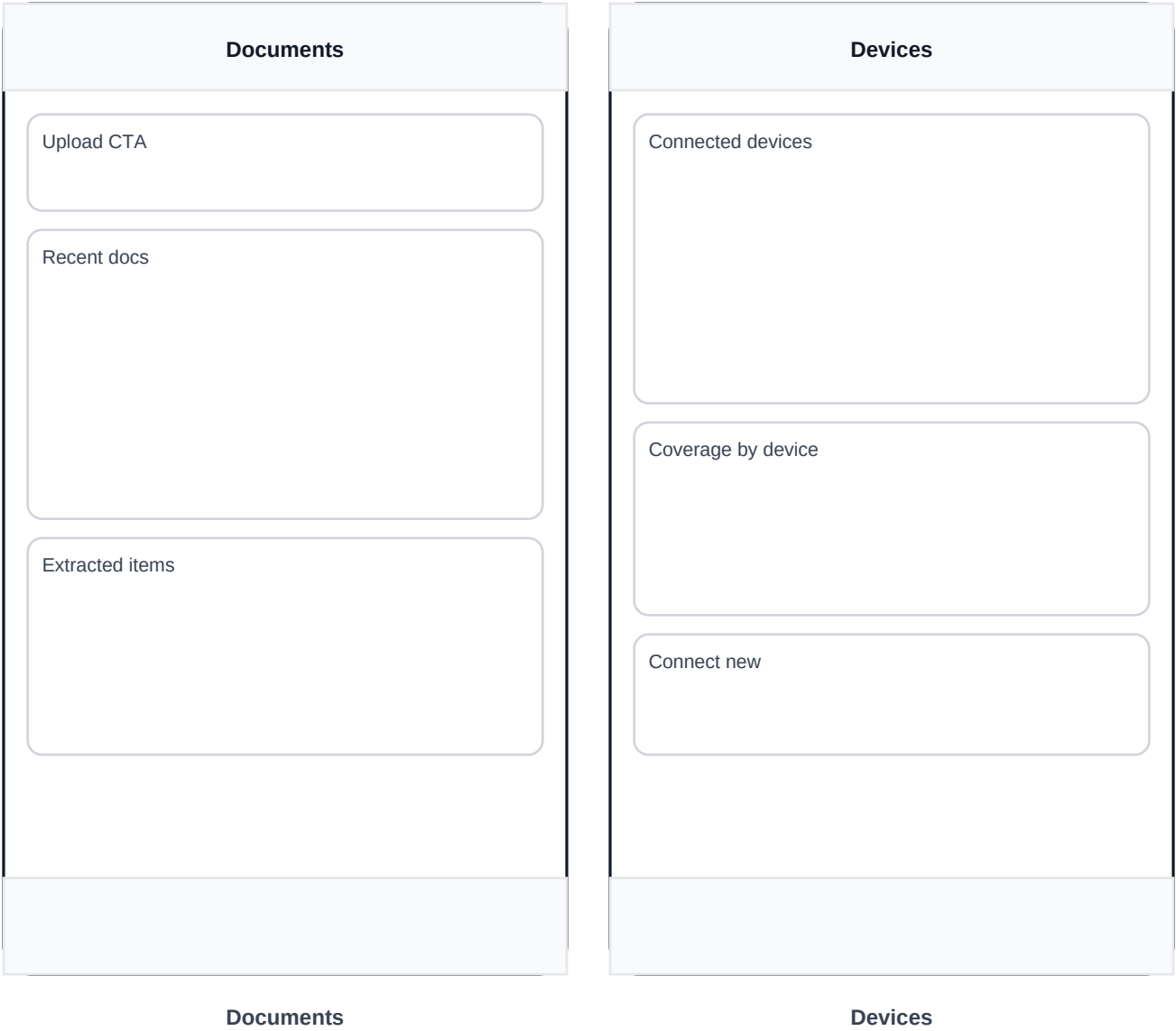
The wireframes below define the **default** layout. Each page supports progressive disclosure: start with a small set of high-signal cards, then expand to details.



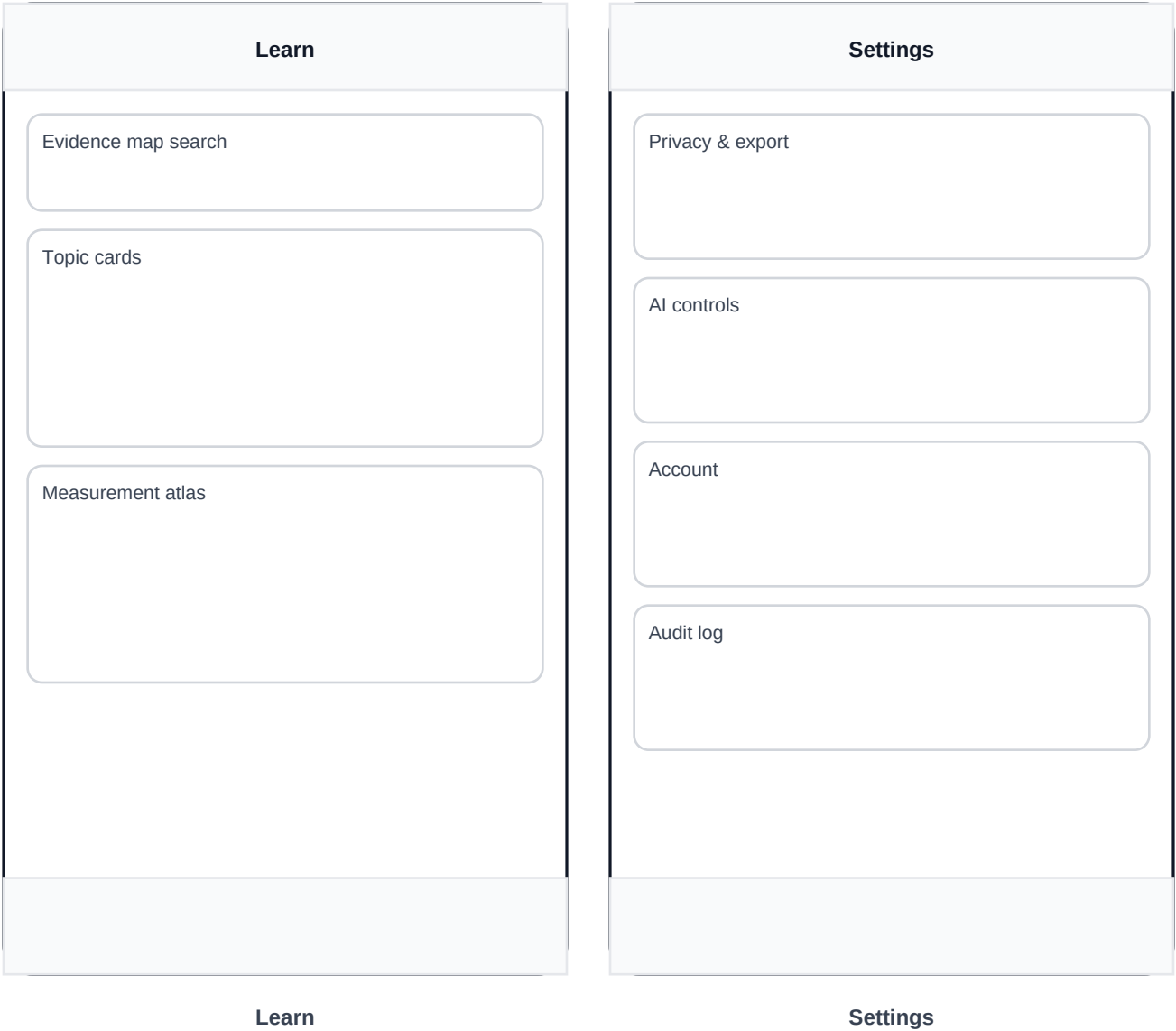
- **Home:** Priority stack: 3–7 cards, sorted by expected impact × urgency × feasibility.
- Weekly snapshot: always visible (sleep/activity/weight/BP/glucose if present).
- Coverage meter: what domains have enough data to personalize recommendations.
- **Timeline:** Single canonical history. All data resolves to events with provenance.
- Filters as chips (labs, meds, symptoms, encounters, imaging, devices).
- Compare mode: pick two dates to see deltas (labs, meds, weight, BP).



- **Insights:** Domain tabs (Cardio, Metabolic, Sleep, Mental, Kidney, Liver/GI, Neuro, MSK).
- Each domain shows trends + variability and drivers for key risk scores.
- Show uncertainty and data quality (coverage, missingness, device bias).
- **Plan:** Recommendations are tasks with measurement plans; never a wall of text.
- Experiment templates (baseline → intervention → re-measure).
- Burden-aware scheduling: don't stack 10 changes at once.



- **Documents:** Document vault with extracted entities preview.
- Side-by-side viewer: original PDF/image + extracted structured data.
- Low-confidence extractions get a Review queue.
- **Devices:** Connection hub; show last sync, permissions, and coverage gaps.
- Per-device limitations microcopy (e.g., PPG vs ECG, sleep stage uncertainty).
- Support multiple devices but designate one as primary per metric.



- **Learn:** Evidence map: intervention → outcomes → effect sizes; show citations and caveats.
- Measurement atlas: invasiveness vs cost vs signal.
- Two explanation modes: plain language and technical.
- **Settings:** Privacy is a product feature: export, delete, audit log, encryption status.
- AI controls: what models are used, what data leaves device, opt-outs.
- Sharing: generate clinician packet links with expirations.

4. Data Ingestion, Normalization, and Provenance

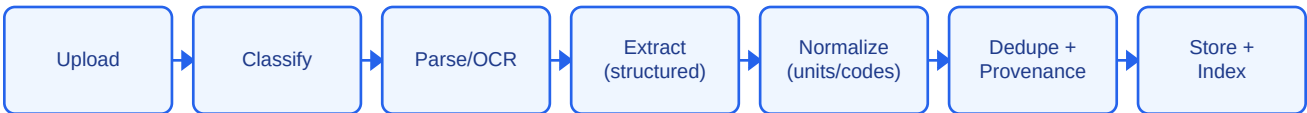
How BioAssist turns messy inputs into a reliable longitudinal health record.

4.1 Supported input sources (MVP → v1 → v2)

- **MVP**: Apple HealthKit / Android Health Connect; PDF/image uploads (labs, visit summaries); manual entry (weight, BP, meds).
- **v1**: major wearables (Fitbit, Oura, Garmin, Withings); CSV import; OTC test photos; receipt ingestion.
- **v2**: EHR via SMART-on-FHIR; lab vendor portals; radiology reports; pharmacy feeds; employer wellness feeds (optional).

4.2 Document ingestion pipeline

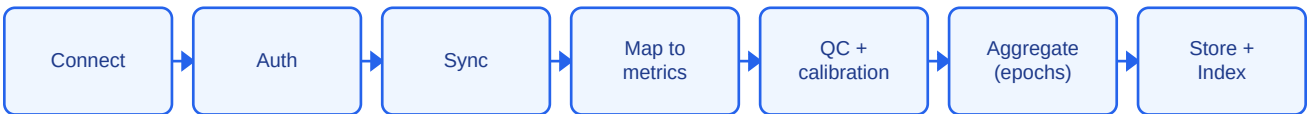
Documents must always remain accessible in their original form. Structured extractions are *derived* and must point back to a source location (page, line/region, timestamp).



Key design choice: provenance as a first-class field

Every extracted datum is stored as: **(value, unit, timestamp, interpretation, source)**. The source includes document id + page + bounding box (for images) or line range (for text), plus extraction model version and confidence.

4.3 Device ingestion pipeline



- Normalize to a canonical metric dictionary (e.g., steps, HR, HRV, sleep duration).
- Store raw time-series when available; always store derived daily aggregates.

- Quality control flags: missingness, implausible values, device swaps, timezone shifts.
- Allow per-metric primary device selection to avoid double counting.

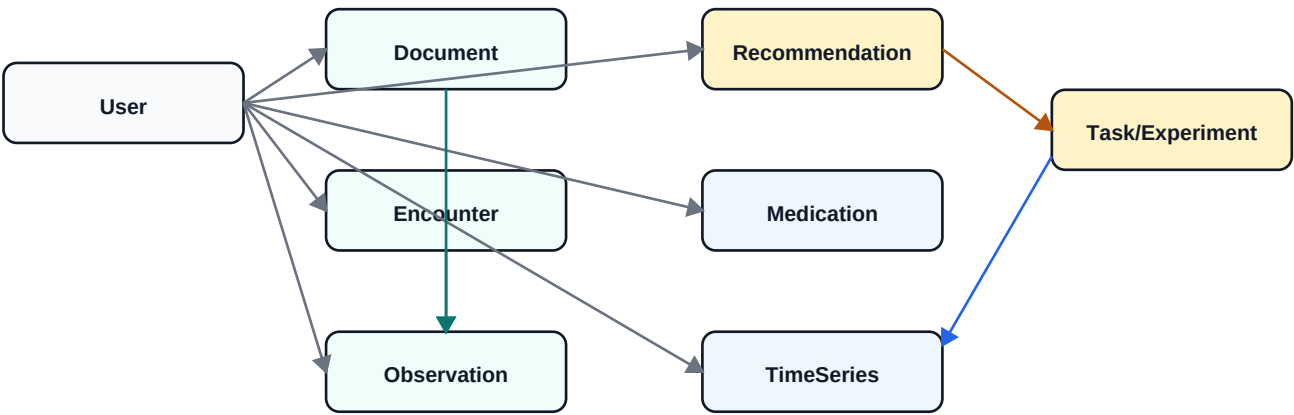
4.4 Normalization rules (non-negotiable)

- **Units:** canonical units + explicit conversion history (e.g., mg/dL ↔ mmol/L).
- **Reference ranges:** store per-lab ranges (they differ) and map to standardized interpretation.
- **Codes:** map to LOINC/SNOMED/RxNorm when possible, but keep original labels.
- **Dedupe:** detect duplicates across PDFs/EHR by (analyte, timestamp, value, provider).
- **Versioning:** store algorithm/model versions for derived metrics and extractions.

5. Canonical Data Model

FHIR-aligned where it helps, plus time-series and provenance primitives that FHIR doesn't model cleanly.

BioAssist should treat the LLM as a *parser* and *explainer*, not as the database. The database stores structured facts with provenance and versioning.



5.1 Core entities (minimum viable)

Field	Type	Notes
document_id	UUID	Stable identifier; used for provenance links.
type	enum	lab_report, visit_note, imaging_report, receipt, etc.
source	enum	upload, EHR, email, device-export, etc.
captured_at	datetime	When the document was created/issued (if known).
uploaded_at	datetime	When user uploaded/linked it.
file_uri	string	Pointer to encrypted object storage.
hash	string	Content hash for dedupe and tamper detection.
extraction_status	enum	pending, extracted, needs_review, failed
extraction_meta	json	model version, confidence summary, warnings

Field	Type	Notes
observation_id	UUID	
code	string	LOINC when possible; preserve original label.
display_name	string	Human-readable label.

value	number/string	Raw value; keep original too.
unit	string	Canonical unit stored alongside original unit.
effective_time	datetime	Time the measurement applies to.
reference_range	json	Low/high + source lab + age/sex if provided.
interpretation	enum	low/normal/high/critical/unknown
provenance	json	document_id + page + bbox/lines + confidence

Field	Type	Notes
series_id	UUID	
metric	enum	steps, hr, hrv, sleep_stage, glucose, etc.
device_id	UUID	Which device produced the data.
sampling	enum	raw, 1s, 1m, 5m, daily, etc.
timezone	string	Store timezone for correct day aggregation.
data_uri	string	Pointer to compressed chunk storage (parquet/zstd).
quality_flags	json	missingness, outliers, device swap detection
derived_daily	json	daily aggregates for fast queries

Field	Type	Notes
recommendation_id	UUID	
title	string	Short and action-oriented.
category	enum	sleep, cardio, metabolic, mental, etc.
trigger	json	Data-driven condition(s) that caused it.
expected_effect	json	Outcome → effect range + uncertainty.
risks	json	Contraindications + downsides + escalation rules.
burden	json	Time/cost/complexity estimate.
measurement_plan	json	What to measure, cadence, stop rules.
evidence_refs	list	Pointers to internal evidence map items.
status	enum	proposed, accepted, snoozed, completed, archived

5.2 Provenance primitive (SourceSpan)

A single, reusable provenance object that can attach to any derived record.

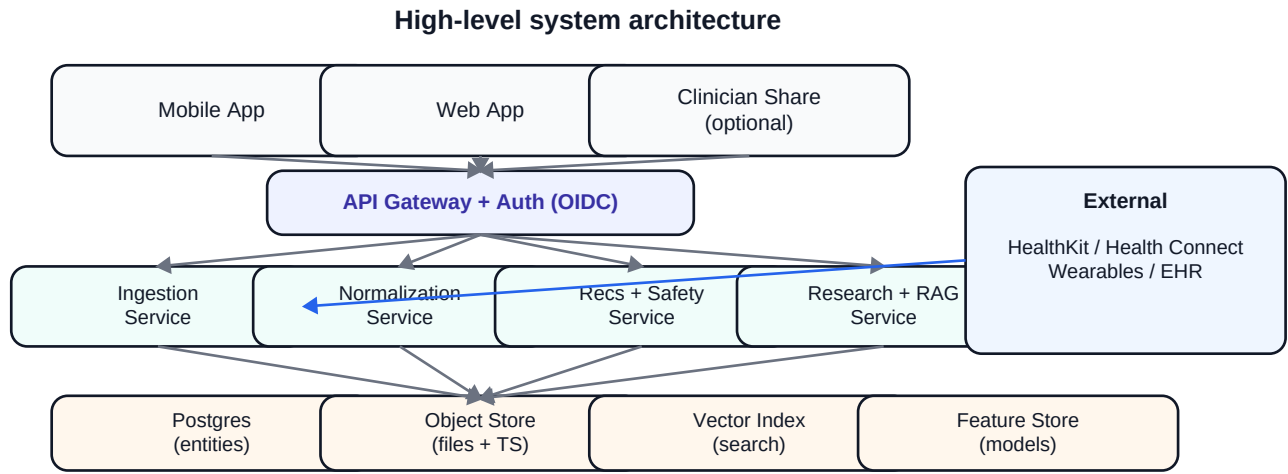
```
SourceSpan = {  
  document_id,  
  page,  
  bbox_or_line_range,  
  extracted_by,  
  model_version,  
  confidence,  
  created_at  
}
```

5.3 Storage choices (pragmatic defaults)

- **Postgres**: entities (users, documents, observations, recs, tasks, permissions).
- **Object storage**: original files + time-series chunks (encrypted at rest).
- **Time-series query layer**: start with daily aggregates in Postgres; add a TSDB when needed.
- **Vector index**: document embeddings for semantic search + RAG over user's corpus (opt-in).
- **Feature store**: derived metrics and risk model outputs with versioning.

6. Architecture and API Surface

Service boundaries, internal contracts, and how data flows end-to-end.



6.1 Services (suggested boundaries)

- **Ingestion Service:** file uploads, device sync jobs, EHR connectors, virus scanning, metadata capture.
- **Normalization Service:** unit conversion, code mapping, reference ranges, dedupe, provenance stitching.
- **Recs + Safety Service:** red-flag detection, guideline reminders, risk scores, recommendation generation.
- **Research + RAG Service:** evidence map, citation retrieval, technical explanations.
- **Search Service** (optional early): index documents + structured entities for global search.

6.2 Internal contracts: events, not tight coupling

Use an event bus / job queue so ingestion, extraction, and recommendation work can be asynchronous and retryable.

- Example events: `document.uploaded`, `document.extracted`, `timeseries.synced`, `observation.created`, `risk.updated`, `recommendation.generated`.
- Every event payload includes **user_id**, **source**, **version**, and **idempotency key**.
- Store job traces so support can debug failed ingestions without asking users to re-upload.

6.3 Public API (user-owned export)

A profitable platform still needs an escape hatch. Exportability is a trust and retention feature.

- Export formats: FHIR bundle (best effort), CSV (labs/vitals), JSON (recommendations/tasks), and original documents.
- Share links: signed URLs with expiration; optional clinician viewer with read-only timeline and key metrics.
- Audit log endpoints: who accessed what, when, and why.

6.4 Observability and quality metrics

- Extraction accuracy metrics: field-level precision/recall on a labeled set of common documents.
- Data freshness: last sync per device; stale data alerts only when it affects recommendations.
- Recommendation quality: acceptance rate, completion rate, “felt helpful” feedback, and adverse-event reports.
- Latency budgets: Home page must load fast for cached summaries; long computations happen server-side.

7. Recommendation Engine and Safety Guardrails

How BioAssist generates advice that is measurable, personalized, and safe.

7.1 Layered recommendation stack

- **Layer 0 — Safety triage:** deterministic red-flag detection with clear “seek care” messaging.
- **Layer 1 — Guidelines:** preventive screening reminders, medication reconciliation prompts, calibration reminders.
- **Layer 2 — Risk models:** validated scores computed deterministically (with driver analysis).
- **Layer 3 — Evidence levers:** map deficits to interventions with expected effect ranges and citations.
- **Layer 4 — Personalization:** filter and rank by constraints (budget/time/preferences/injuries).
- **Layer 5 — Experimentation:** N-of-1 protocols with stop rules and re-measurement.

7.2 Recommendation contract (canonical JSON schema)

This schema is the foundation of explainability, auditing, and product quality. It also enables paywalled advanced recommendations without breaking UX consistency.

```
{
  "recommendation_id": "uuid",
  "title": "string",
  "category": "sleep|cardio|metabolic|mental|nutrition|exercise|preventive|other",
  "priority": { "impact": 0-100, "urgency": 0-100, "feasibility": 0-100, "overall": 0-100 },
  "trigger": {
    "signals": [
      { "type": "observation|timeseries|survey|document",
        "id": "uuid",
        "field": "e.g., A1c, LDL, sleep_duration",
        "value": "number|string",
        "unit": "string",
        "time_window": "e.g., last 30d",
        "provenance": { "source_span_id": "uuid", "confidence": 0-1.0 }
      }
    ],
    "logic": "human-readable explanation of why it fired"
  },
  "expected_effect": [
    { "outcome": "e.g., systolic_bp, mood_score, all_cause_mortality",
      "effect_range": { "low": "value", "high": "value" },
      "time_horizon": "e.g., 8-12 weeks",
      "uncertainty_notes": "string"
    }
  ],
  "risks_and_constraints": {
    "contraindications": ["string"],
    "tradeoffs": ["string"],
    "escalation": ["when to seek care / urgent flags"]
  },
  "burden": { "time_min_per_day": 0, "cost_estimate_usd": 0, "complexity": "low|med|high" },
  "steps": [
    { "label": "action step", "frequency": "once|daily|weekly", "how_to": "short" }
  ],
  "measurement_plan": [
    { "metric": "string", "method": "device|lab|survey|manual", "cadence": "string", "stop_rules": ["string"]
    },
  ],
  "evidence_refs": [
    { "key": "internal_evidence_id", "grade": "A|B|C|D", "summary": "string" }
  ],
  "status": "proposed|accepted|snoozed|completed|archived"
}
```

7.3 Ranking and “priority stack” algorithm

Home should show the smallest set of actions that maximizes impact while minimizing burden.

- Compute **overall priority** = Impact × Urgency × Feasibility (scaled 0–100).
- Impact: expected change in the user’s selected outcomes (longevity, energy, mood, performance).
- Urgency: safety risk, time-sensitive screenings, rapidly worsening trends.
- Feasibility: user constraints + friction estimate + past adherence.
- Always include a quick win item (≤ 2 minutes) when possible (e.g., connect device, confirm med list).

7.4 Safety guardrails (minimum set)

- No medication dosing changes. Medication-related output is limited to reconciliation, adherence, and “discuss with clinician” suggestions.
- For abnormal labs or serious symptoms: use neutral language; avoid deterministic diagnoses.
- Display uncertainty: confidence, data quality, and limitations (device bias, missing data).
- Escalation paths: urgent vs non-urgent messaging; emergency guidance remains generic.
- Human-in-the-loop option (paid add-on): clinician/nurse review of flagged items.

7.5 Starter recommendation library (first release)

Start with high-value, low-risk recommendations that rely on common signals and close the measurement loop.

Sleep

- Set a consistent wake time (+/- 30 minutes) for 2 weeks; measure sleep duration + daytime sleepiness.
- If snoring / witnessed apneas / high sleepiness: complete sleep risk screener; consider HSAT referral discussion.
- Reduce late caffeine (none after a chosen cutoff); track sleep onset latency and perceived quality.

Cardiometabolic

- Baseline home blood pressure protocol (validated cuff; 2 readings AM/PM for 7 days).
- If BMI/waist trends upward: establish weekly weight + waist trend line; choose one nutrition lever to test.
- If elevated fasting glucose/A1c: add post-meal walk experiment; measure glucose (if available) or weight/energy.

Exercise & Fitness

- Establish zone-2 cardio baseline (minutes/week) and progress by 5–10%/week.
- 2–3x/week strength training minimum effective dose with progressive overload tracking.
- VO2 estimate baseline (wearable or field test); retest every 8–12 weeks.

Mental Health

- Weekly mood + stress check-in (PHQ-2/GAD-2 style screeners) + sleep correlation review.
- Social connection goal (2 meaningful contacts/week); track loneliness score.
- If persistent low mood/anxiety: present evidence-based options (therapy, sleep, exercise) and encourage clinician discussion.

Preventive & Admin

- Generate a clinician packet before appointments: timeline highlights + meds + questions list.
- Medication list reconciliation monthly; identify duplicates/interactions (prompt clinician/pharmacist review).
- Vaccination/age-appropriate screening reminders (rules engine; user can confirm completed).

8. Evidence Map and Research System

How BioAssist stays science-based without hallucinating.

8.1 Evidence map data model

- **Intervention:** sleep regularity, resistance training, DASH-style diet pattern, etc.
- **Outcome:** BP, A1c, LDL, mortality, mood score, VO2max, etc.
- **Population:** age range, sex, baseline risk, condition-specific cohorts.
- **Effect size:** range + confidence intervals when available; time horizon.
- **Quality grade:** simple rubric (A–D) based on study design + consistency.
- **Harms/downsides:** adverse events, burden, contraindications.
- **Citations:** canonical IDs + links; store retrieval date and source type.

8.2 Research pipeline (human-reviewed)

Automate the boring parts, but keep review loops for anything that changes recommendations.

- Automated search + triage: retrieve candidate papers/guidelines; dedupe; tag by domain/outcome.
- Structured extraction: population, intervention details, outcomes, effect sizes, limitations.
- Human review queue: approve evidence entries and assign grades; track reviewer + date.
- Change management: when evidence changes, recompute affected recommendation templates and flag users whose plans reference them.

8.3 “No-citation, no-claim” rule for scientific statements

Policy

Any statement about expected outcomes (e.g., “X lowers blood pressure”) must reference an evidence entry. If the system cannot cite, it must label the statement as a hypothesis or omit it.

8.4 Two explanation modes

- **Plain:** what to do, why it matters, what to measure, common pitfalls.
- **Technical:** mechanism, effect sizes, confidence intervals, study types, heterogeneity notes.

9. Integrations Roadmap

Connect the highest-value sources first, with the least engineering and compliance drag.

9.1 Integration philosophy

- Prefer **aggregators** (HealthKit/Health Connect) over direct device APIs early.
- Ingest **documents** early: users already have PDFs; it's immediate depth.
- Design all connectors as **adapters** that map into the same canonical metric dictionary.
- Show limitations in-product (e.g., sleep stage uncertainty, PPG artifacts).

9.2 Priority matrix

Integration	User Value	Complexity	Notes
Apple HealthKit	High	Low–Med	Fastest path to steps, HR, sleep, workouts.
Android Health Connect	High	Low–Med	Android equivalent aggregation layer.
PDF/Image upload	High	Low	Unlocks clinical depth immediately.
Fitbit / Oura / Garmin	Med	Med	Improves sleep/HRV/activity fidelity for many users.
Withings (scale/BP)	Med	Med	High-signal home vitals and weight trend.
Dexcom / Abbott CGM	Med	High	Powerful metabolic signal; fewer users; careful UX.
Labs portals (Quest/Labcorp etc.)	Med	High	Convenience; messy mapping; can be phased.
SMART-on-FHIR (EHR)	High	High	Big value, high variability; do after stable core.
Pharmacy feeds	Med	High	Medication list accuracy; compliance heavy.

9.3 Device connection UX requirements

- Single Connect button per integration with clear permission copy (what data, why).
- Post-connection: show last sync, coverage, and what's missing.
- Allow user to choose which device is primary for each metric; avoid double counting.
- When the user changes devices, handle discontinuities and explain why trends may shift.

10. Privacy, Security, and Trust

Trust is a growth lever in health. Build it into the UI and the architecture.

10.1 Privacy UX primitives

- **Data ledger:** show all connected sources, permissions, and last access.
- **Export:** one-click export (structured + originals).
- **Delete:** clear deletion pathways with confirmation and time windows.
- **Sharing:** clinician packet links with expiration and scope controls (read-only).
- **AI transparency:** show what model ran and what data it used for each explanation (high level).

10.2 Security architecture (baseline)

- Encryption in transit (TLS) and at rest (per-tenant keys where feasible).
- Separate object storage for documents with strict IAM; signed URLs only.
- Role-based access control + least privilege for services.
- Immutable audit log for PHI access (user + support + system jobs).
- Threat model for: account takeover, link sharing, prompt injection from uploads, data exfiltration.

10.3 Compliance posture (practical)

Exact requirements depend on whether you operate as a consumer app, covered entity associate, or both. Plan for HIPAA-grade controls early if you intend to integrate with EHRs.

- Maintain policies for data retention, incident response, access reviews, and vendor management.
- SOC 2 controls map cleanly to health trust; start collecting evidence early.
- If offering clinician services, treat it as a separate, well-scoped workflow with clear consent.

Important product copy (suggested)

BioAssist provides information and tools to help you understand and organize health data. It is not a substitute for professional medical advice, diagnosis, or treatment.

11. Monetization and Business Model

Make the product profitable without corrupting trust or clinical safety.

11.1 Value-based packaging (tiers)

Plan	Price (example)	Included
Free	0	Timeline, basic dashboards, limited uploads, HealthKit/Health Connect, basic recommendations, export (structured).
Pro	19–39/mo	Unlimited uploads, advanced insights, risk score drivers, experiment mode, evidence map, premium device integrations, automations (weekly report).
Family	29–59/mo	Multiple profiles, caregiver mode, shared clinician packets, family reminders.
Clinician Review Add-on	49–199/mo	Human review of flagged items, monthly packet, Q&A; (scope-limited).
Enterprise / Clinic	Custom	White-labeled patient packets, API access, SSO, admin dashboards, compliance contracts.

11.2 Paywall philosophy

- Do not paywall data access (timeline, export). Paywall processing (advanced insights, premium recs, automation).
- Use unlock moments: show a preview of what Pro adds without blocking basics.
- Prefer feature-based limits over nagging usage limits (health is sensitive).
- Premium upsell should feel like: less work, more clarity — not fear.

11.3 Revenue levers beyond subscriptions

- Clinician packet generation + secure sharing (paid add-on for heavy users).
- Lab test marketplace / scheduling (if done, disclose incentives; avoid conflicts).
- Device partnerships and affiliate programs (must be transparent and optional).
- B2B: employer wellness, clinics, longevity practices (requires higher compliance).

11.4 Unit economics checklist

- Big costs: LLM extraction + summarization, storage of originals, device API quotas, support.
- Mitigations: cache derived summaries, batch processing, user-controlled deep analysis toggles.
- Paid tiers can subsidize high-cost workflows (e.g., full document semantic indexing).

12. Growth, Retention, and Engagement Loops

Health products win by being useful every week, not by being addictive.

12.1 Retention loops (recommended)

- **Weekly recap:** 1-page summary of trends + 1–3 suggested actions + what to measure next.
- **Appointment prep:** when the user adds an upcoming visit date, prompt packet creation 48–72h before.
- **Coverage ladder:** show next best connection (e.g., add BP cuff) only when it unlocks meaningful insights.
- **Experiment check-ins:** gentle prompts to re-measure at the planned horizon.

12.2 Acquisition channels (practical)

- SEO + content: evidence-based explainers tied to your measurement atlas (high intent).
- Referrals: share your clinician packet and share your trend chart (with privacy controls).
- Partnerships: longevity clinics, personal trainers, dietitians, therapists (tool not replacement).
- Product-led: make exports and packet sharing watermarked with bioassist.ai (opt-in).

12.3 Avoid these engagement anti-patterns

- Fear-based notifications (over-alerting)
- Leaderboard gamification for sensitive health metrics
- Opaque AI advice without sources or measurement plans

13. Roadmap and Build Plan

A pragmatic sequence that gets to revenue fast without sacrificing core correctness.

13.1 Milestone roadmap

Time	Theme	Deliverables
0–30 days	Foundations	Auth, user model, document vault, basic upload, HealthKit/Health Connect connect, timeline v0, export v0.
31–60 days	Normalization + Insights v1	Structured extraction for common lab PDFs; units/reference ranges; weekly snapshot; basic trend charts.
61–90 days	Plan v1 + Safety v1	Recommendation contract, priority stack, templates, red-flag engine v1, clinician packet v0.
3–6 months	Integrations + Pro tier	Major wearable connectors, receipt parsing, evidence map UI, experiments, subscription + paywall.
6–12 months	EHR + Scale	SMART-on-FHIR, deeper risk models, enterprise/clinic pilots, compliance upgrades, human review add-on.

13.2 MVP definition (must ship)

- **Home:** priority stack + weekly snapshot + coverage meter.
- **Timeline:** labs + device summaries + documents, with provenance badges.
- **Documents:** upload PDFs/images; extraction for common labs; review queue for low confidence.
- **Plan:** 10–20 starter recommendations with measurement plans; task tracking.
- **Privacy:** export + delete + audit log basics.

13.3 Team and responsibilities (minimum viable team)

- 1 Product/Design lead (IA, UX, copy, research workflows)
- 2–3 Full-stack engineers (web/mobile + API)
- 1 Data/ML engineer (extraction, normalization, evaluation harness)
- 1 Clinical advisor (safety, guideline sanity, escalation logic)
- 1 Growth/Content (optional early)

13.4 Quality gates

- Extraction tests: gold set of ~200 documents across labs/notes; field-level accuracy targets.
- Safety tests: synthetic cases to ensure correct escalation wording and no dangerous outputs.
- UX tests: onboarding time-to-value < 2 minutes for device connect or upload flow.
- Performance: cached home loads fast; background jobs retry safely.

14. Analytics and Product Instrumentation

Measure value delivered, not just engagement.

14.1 Key product metrics

- **Activation:** connected device or uploaded document + saw first insight within 24h.
- **Weekly active users** and weekly report opens.
- **Actionable Signal:** accepted recommendations per week × completion rate.
- **Data coverage:** percent of key metrics populated for a user’s selected goals.
- **Trust:** export usage, privacy settings engagement, support ticket sentiment.

14.2 Event taxonomy (starter set)

Event	Meaning
onboarding.started	User begins onboarding flow.
integration.connected	User connects a device/EHR integration.
document.uploaded	Document uploaded; store type and page count.
document.extraction.reviewed	User corrected an extraction; capture fields corrected.
home.recommendation.viewed	A recommendation card was opened.
recommendation.accepted	User accepts a recommendation into their plan.
task.completed	Plan task completed.
packet.generated	Clinician packet exported/shared.
subscription.started	Paid conversion.
subscription.churned	Cancellation + reason.

14.3 Feedback loops

- Thumbs up/down on recommendations with why tags (too hard, not relevant, unclear, etc.).
- Post-extraction review: “Was this accurate?” to improve parsing.
- Lightweight weekly NPS prompt (rare; only after value moments).

15. AI/LLM Implementation Notes

How to use LLMs for extraction and explanation without turning them into a source of truth.

15.1 Where LLMs are appropriate

- **Document classification** (type detection)
- **Structured extraction** into validated schemas (labs, meds, problems, procedures)
- **Summarization** of a document into a short clinician-friendly note
- **Conversational layer** over the user's own data with citations to sources
- **Evidence synthesis** with citations and human review

15.2 Where LLMs are not appropriate (or must be heavily constrained)

- Uncited medical claims or diagnoses
- Medication dosing advice or changes
- Automated escalation decisions without deterministic rules
- Replacing structured storage (LLM as database)

15.3 Prompting patterns (recommended)

- Use **strict JSON schema outputs**; reject and retry if invalid.
- Retrieve context from the user's data store; never ask the model to remember.
- For Q&A;: response must include source links (document ids / timeline events).
- For summarization: include provenance counters (how many docs included, what time range).

15.4 Prompt injection and document safety

- Treat uploads as untrusted input; never allow documents to modify system prompts.
- Strip active content; scan for malware; isolate rendering.
- When running RAG: only retrieve from whitelisted stores (user docs + curated evidence).
- Rate-limit and log AI actions; provide user-facing explanation for why we need this data.

16. Appendix: API Surface and Contracts (Starter)

A minimal set of endpoints and contracts to support the MVP UI.

This is a starter outline. The core requirement is consistency: every UI surface should be backed by stable, cacheable APIs and structured resources.

16.1 Example REST endpoints

Endpoint	Purpose
POST /v1/documents/upload	Upload PDF/image; returns document_id; starts extraction job.
GET /v1/timeline?from=&to;=	Paginated timeline events with provenance links.
GET /v1/observations?code=&window;=	Structured observations for a code/time window.
GET /v1/metrics/daily?metric=steps	Daily aggregates for a metric.
GET /v1/recommendations	Current recommendation stack (ranked).
POST /v1/recommendations/{id}/accept	Accept into plan; generates tasks.
GET /v1/plan	Tasks, experiments, schedules.
POST /v1/exports	Generate export bundle; returns signed URL.
GET /v1/auditlog	User-visible audit log entries.

16.2 UI query model (recommended pattern)

- Home uses one aggregated endpoint: GET /v1/home returning snapshot + priority stack + coverage.
- Timeline uses cursor pagination; include summary cards for dense time-series (e.g., sleep week summary).
- Documents list endpoint returns both originals and extraction status; viewer endpoint returns extracted entities + provenance.

16.3 Caching strategy

- Cache read models (home snapshot, weekly summaries) per user with short TTL.
- Invalidate caches on new observations, new time-series sync, or recommendation updates.
- Store derived summaries with version keys so model upgrades don't corrupt history.

17. Build Checklist

Concrete epics to move from spec to shipped product.

17.1 UX component checklist (Material-inspired)

- Top app bar with search, profile, and sync status indicator
- Bottom nav / sidebar with consistent labels and icons
- Card component with: title, subtitle, metric chips, provenance badge, and primary action button
- Chart component with: time range selector, annotation support, and data quality overlay
- Upload flow: drag/drop (web), camera scan (mobile), multi-file batching
- Review flow for low-confidence extractions (diff view + approve/edit)
- Recommendation card with Why/Risks/Measure/Accept/Snooze actions
- Empty states with one-click next action

17.2 Backend epics

- **Identity + consent:** OIDC login, device permissions, sharing consent.
- **Document vault:** encrypted storage, virus scanning, metadata, hashing.
- **Extraction + review:** schema validation, confidence, review queue UI.
- **Normalization:** units, codes, reference ranges, dedupe, versioning.
- **Time-series ingestion:** HealthKit/Health Connect ingestion + daily aggregation.
- **Recommendations:** priority stack endpoint, task generation, history tracking.
- **Export + audit:** export bundle generator, signed URLs, audit log store.

17.3 Launch checklist

- Privacy policy + terms, plain-language in-app explanations.
- Incident response runbook and support process.
- Baseline evaluation set for extraction + rec safety.
- App store review readiness (if mobile) + permissions copy that passes review.

End of document.